OUTCOMES-BASED EDUCATION (OBE) COURSE SYLLABUS IN
CC102 – COMPUTER PROGRAMMING 1
Date revised/enhanced: AUGUST 2025

• **School Year/Semester**	AY 2025-2026/ 1st Semester
• **University**	Vision: A globally-recognized green research University committed to excellent service to the nation. Mission Foster excellence, holistic and inclusive education attuned to the demands of a diverse world towards the formation of productive and value-laden lives. Goal Uphold the tradition of excellence in instruction, research, extension and production geared towards a strong academe-industry interface for societal development. Core Values Respect, Integrity, Social Responsibility, Excellence, Commitment Institutional Outcomes: • A *Professiona*l who is morally upright, socially responsible and globally employable. • A *Leader and Innovator* who inspires others and is committed to social transformation and national development. • An *Environmental Advocate* committed to research, extension and production initiatives.
• **College/Campus**	College of Information and Communications Technology/ Main Campus
• **Program/Degree**	Bachelor of Science in Information Technology (BSInfoTech.)

• **Program Outcomes**

• Articulate and discuss the latest developments in the specific field of practice.	**P**
• Effectively communicate orally and in writing using both English and Filipino.	**P**
• Work effectively and independently in multi-disciplinary and multi-cultural teams.	**P**
• Act in recognition of professional, social and ethical responsibility.	**P**
• Preserve and promote “Filipino historical and cultural heritage”.	**P**
• Apply knowledge of computing science, and mathematics appropriate to the discipline.	**P**
• Understand best practices and standards and their applications.	**I**
• Analyze complex problems and identify and define the computing requirements appropriate to its solution.	**P**
• Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.	**I**
• Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.	**I**
• Integrate IT-based solutions into the user environment effectively.	**P**
• Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession.	**P**
• Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.	**I**
• Assist in the creation of an effective IT project plan.	**I**
• Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.	**I**
• Analyze the local and global impact of computing information technology on individuals, organizations, and society.	**I**
• Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.	**I**
• Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development.	**I**

	• Participate in the generation of new knowledge or in research and development projects	**I**
	• Support local, regional and national development plans along education, environment, socio economic, health, gender and development, science and technology.	**I**

• **Course Code/Title**	**CC102 – COMPUTER PROGRAMMING 1**
• **Course Description**	The course covers the use of general-purpose programming language to solve problems. The emphasis is to train students to design, implement, test, and debug programs intended to solve computing problems using fundamental programming constructs.
• **Course Credit**	3 units (2 units lecture, 1-unit laboratory)
• **Prerequisite**	None
• **Contact Hours**	5 hours / week (2.5 hours lecture, 2.5 hours laboratory)

• **Course Outcomes**

At the end of the course, the students shall be able to:

COs	Description	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
CO1	Apply C programming language syntax, semantics, and standard libraries to solve a variety of computational problems.	**2**	**1**	**1**	**1**	**1**	**3**	**3**	**3**	**3**	**3**	**3**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
CO2	Demonstrate a solid understanding of fundamental programming concepts, including algorithms, data structures, and control flow.	**2**	**1**	**1**	**1**	**1**	**3**	**3**	**3**	**3**	**3**	**3**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
CO3	Design and implement efficient and well-structured C programs using modularization and functions.	**2**	**1**	**1**	**1**	**1**	**3**	**3**	**3**	**3**	**3**	**3**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
CO4	Utilize arrays and other data structures to effectively represent and manipulate data within C programs.	**2**	**1**	**1**	**1**	**1**	**3**	**3**	**3**	**3**	**3**	**3**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
					1 – CO has minor contributions to PO **2** – CO has a moderate contribution to PO **3** – CO has major contributions to PO																	

• **Course Outline/Learning Plan**

POs	COs	Desired Learning Outcomes	Course Content/ Subject Matter	Textbooks/ References	Teaching and Learning Activities *(TLAs)*	Assessment Tasks	Instructional Resources/ Materials	Time Table (Hours)
A F G H I L N	**CO1**	At the end of the semester, the students shall be able to: • Analyze fundamental programming concepts and apply them to construct basic C programs. • Synthesize knowledge of C syntax and structure to effectively represent problem solutions in code.	• **Introduction to Programming and C** • Basic Programming Concepts • Algorithms • Flowchart • Pseudocode • Introduction to C • Structure • Compilation • Structure • Core Syntax • Datatypes • Variables • Operators • Input/output	Slobodan Dmitrović (2024) Modern c for Absolute Beginners. Apress, Jackiewicz, Rafal (2023) *Practical C*. Independently	• Lecture • Interactive discussion\ • Blended/Hybrid Learning	• Quizzes • Assignment Activities	• Handouts/ Module • Presentation	22.5
A F G H I L N	**CO3**	• Effectively utilize operators to manipulate data and perform calculations within C programs. • Apply decision-making and looping	• **Operators Control Structures** • Operators • Decision Making Statements • Looping Statements	Slobodan Dmitrović (2024) Jackiewicz, Rafal (2023)	• Lecture • Interactive discussion • Guided Hands-on Programming Sessions • Blended/Hybrid Learning	• Quizzes • Assignment • Activities	• Handouts/ Module • Presentation	20

		constructs to control program flow and create dynamic program behavior.						
MID-TERM EXAMINATION 2.5								
A F G H I L N	**CO2 CO3 CO4**	• Employ operators and expressions to construct complex computations and logical evaluations within C programs. • Design and implement effective control flow structures to guide program execution based on specified conditions and iterative processes.	• **Functions** • Function definition, declaration, and calling • Function parameters and return values • Function prototypes	Slobodan Dmitrović (2024)	• Lecture • Brainstorming • Interactive discussion • Blended/Hybrid Learning	• Seatwork Activity • Assignment	• Computer/ Laptop • Internet Connection	22.5
A F G H I L N	**CO2 CO3 CO4**	• Utilize arrays to store and manipulate collections of data, demonstrating proficiency in indexing, assignment, and traversal techniques. Apply the concept of multidimensional arrays to represent and process complex data structures, enhancing problem-solving capabilities.	• **Arrays** • Array declaration and initialization • Array access and manipulation • Introduction to Multidimensional arrays	Slobodan Dmitrović (2024)	• Interactive discussion • Programming Session • Blended/Hybrid Learning	• Project Presentation	• Computer/ Laptop • Internet Connection	20
FINAL EXAMINATION 2.5								

• **Suggested Readings and References**	• W3Schools.com. (2024). W3schools.com. https://www.w3schools.com/c/c_intro.php • Learn C Programming. (2023). Programiz.com. https://www.programiz.com/c-programming • GeeksforGeeks. (2024, June 10). C Programming Language Tutorial. GeeksforGeeks; GeeksforGeeks. https://www.geeksforgeeks.org/c-programming-language/ • S, R. A. (2021, November 26). Getting Started with C Programming: A Beginner's Guide! Simplilearn.com; Simplilearn. https://www.simplilearn.com/tutorials/c-tutorial/use-of-c-language • Programiz. (2021). #1: Getting Started with C Programming | C Programming for Beginners [YouTube Video]. In YouTube. https://www.youtube.com/watch?v=KnvbUiSxvbM&list=PL98qAXLA6aftD9ZlnjpLhdQAOFI8xIB6e
• **Course Requirements**	• Written outputs in this course such as assignments, quizzes, seatwork, and problem set or exercises, must be submitted before the end of midterm and final period or a written/email formal request must be requested and approved from the Instructor/Professor before the due date. • Take Midterm and Final examinations at the campus. Students are required to submit their individual projects before the commencement of the final examination period.
• **Course Policies**	• Wearing of mask and observance of social distancing shall be imposed when need arises. • Use of cellphones, earphones, and other gadgets that are not necessary for the instruction is not allowed. • Students accumulating more than five (5) absences in the course will be subject to course dropping. • Cheating and plagiarism are strictly prohibited. Students caught doing these acts shall be dealt with in accordance with the provisions in the Student Handbook. • Students found to have used unauthorized AI tools in completing course requirements will receive a grade of 3.0. • Students who arrive thirty (30) minutes or more after the scheduled start of class will be considered absent and will not be permitted to enter the classroom/laboratory. • All course requirements such as assignments, quizzes, seatwork, and problem set, or exercises must be submitted prior to the commencement of the midterm for the midterm period and final examinations for the final period. • Students who miss quizzes or examinations may be considered for alternative arrangements upon presentation of a duly signed excuse letter accompanied by supporting documentation. • In the event of unexpected circumstances, emergencies, or calamities that necessitate a shift to an online or modular learning modality, a blended online platform will be implemented to sustain instructional delivery and academic operations. In cases where class suspensions are declared, alternative instructional strategies, including asynchronous learning and/or self-paced modules, will be provided to ensure continuity. Additionally, provisions will be made for students who are unable to attend online classes due to emergencies, allowing for flexible catch-up mechanisms and special considerations to support their learning progress.
• **Grading System** **Per Board Resolution No. 3,4 & 5, s. 2016**	Distribution of Weight for the Overall/Final Grade Midterm Grade 50% Tentative 50% Distribution of Weight for the Overall/Final Grade Midterm Grade 50% Tentative 50%

	Midterm/Final Examination 30% Quizzes/Homeworks/Seatworks/Problem Sets 30% Performance(skills-based; psychomotor) 40% Total 100%
• **Consultation Time/Venue**	MRBT : MTH 8:00 – 12:00 | via Online/Messenger

Prepared by:

MARY ROSE B. TABLO
Instructor I
Date: _____

Reviewed by: **ASTER VIVIEN C. VARGAS, MSIT** Department Chairperson, Computing Programs Date: _____	Approved: **MARIA CONCEPCION S. VERA, DIT** Dean, CICT Date: _____